

topics, such as plating processes, managing manufacturing variations, leveraging computer-aided manufacturing (CAM), and integrating best practices for enhanced reliability and manufacturability.

Selecting the right CAD tool. The market offers a broad range of CAD tools for designing PCBs. Choosing the right one is crucial for ‘designing for reality.’ While seasoned designers prefer established tools, beginners often gravitate towards free or simpler electronic design automation (EDA) software. Avoid letting managerial preferences dictate the decision. Instead, prioritise software that aligns with your design-to-manufacturing workflow. Selecting an unsuitable CAD tool can lead to three significant issues:

1. Increased design time for complex boards
2. Difficulty achieving error-free designs, especially for high-speed or RF boards
3. Challenges in meeting electrical performance standards for intricate designs

Stay focused on the basics when evaluating design tools. Key questions to consider:

- Is it easy to learn?
- Does it have a user-friendly interface?
- Does it offer features tailored to your design needs?
- Does it include parts libraries or supports custom parts databases?
- Can it generate Gerber files compatible with most manufacturers?
- Does it automate design rule checks?
- Can it produce intelligent file formats like IPC or ODB ++?

Qualifying PCB suppliers for first-attempt success. The major categories identified as the key factors for manufacturer selection:

- Raw materials used for laminates

- Machine capabilities
- Finishes for solder masks (e.g., glossy or matte)
- Delivery schedule flexibility
- Support for controlled impedance
- Types of via filling
- Solder mask colour and legend options
- Special profiling options (e.g., edge finger plating)
- Post-fabrication test methods

For high density interconnect (HDI) designs.

- Select vendors with proven HDI expertise to ensure high yield
- Review manufacturing capabilities early in the design process
- Confirm material availability to avoid delays; any later design change based on MFG review will cause additional timeline issues
- Address special requirements like stack-up selection and broad-side impedance

PCB placement: Do’s and Don’ts

Component placement is where the art and science of PCB design converge. While no single approach works universally, orientation, placement, and layout heavily influence manufacturability.

Do’s.

- Orient similar components to facilitate efficient and error-free assembly (manual + automated)
- Organise through-hole and surface-mount components to minimise assembly steps
- Ensure power, ground, and signal traces have clear paths
- Avoid components prone to heating issues during operation
- Place decoupling capacitors near their respective pins; use the schematic to drive this placement strategy
- Work one major component at a time, surrounding it by its related small parts

- Develop a placement strategy using a floor diagram before starting
- Adhere to mechanical constraints, such as height and keep-out zones

• Try to keep all parts on one side of the board. But if you must place parts on both sides, keep bottom side surface mount parts away from through-hole pins (This will make two-sided assembly possible, if not particularly cost-effective)

• Don’t break the rules or violate the DfX requirement of your supplier. Try to follow standards like IPC recommendation fairly

Don’ts. Here are some tips or no-good items that will help you get started:

- Avoid placing components on the solder side behind through-hole components
- Do not over-rely on auto-routing tools
- Become familiar with design for assembly (DFA) and guidelines provided by supplier, but do not assume the DFA guidelines are static—adapt as needed
- Familiarise yourself with component types, tolerances, and requirements that are unique to placement of PCBs.
- Use a routing grid that supports trace/space rules for consistent patterns

Most importantly, don’t hesitate to ask for help from your manufacturer’s CAM department or customer service. Seek early input from your manufacturer’s CAM department to minimise iterations.

Mastering PCB design is an iterative process requiring technical proficiency, collaboration with reliable manufacturing partners, and an unwavering commitment to precision. By implementing these strategies, designers can consistently produce innovative, manufacturable designs that meet performance and cost objectives. **EFY**